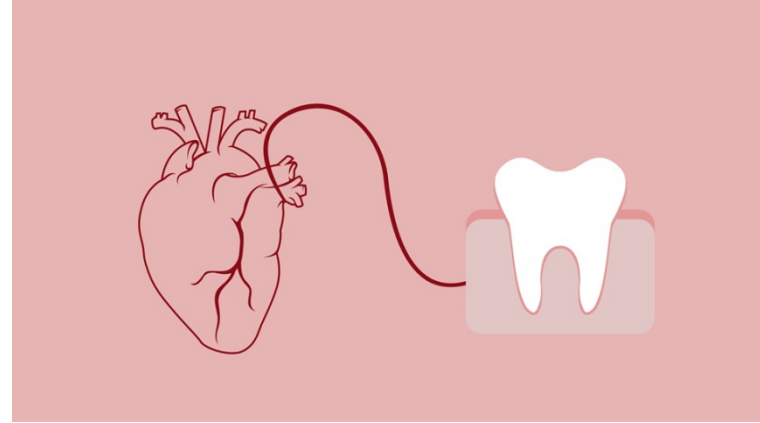
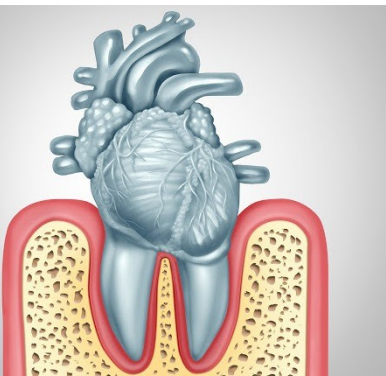




Acute Rheumatic Fever and Rheumatic Heart Disease

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Learning outcomes

- **By the end of lecture, the student will be able to:**
 - **Define acute rheumatic fever (ARF).**
 - **List etiology of ARF.**
 - **Enumerate clinical manifestation of ARF.**
 - **Identify how to diagnose ARF.**
 - **Describe management of ARF.**
 - **Describe rheumatic valvular lesions.**



Acute Rheumatic Fever (ARF)



- It is an immune-mediated inflammatory disorder caused by group A streptococci, 2-4 weeks following to upper respiratory tract infection as pharyngitis or skin infections. It affects the heart, skin, joints and CNS.
- It characterized mainly by:
 - Polyarthrititis, chorea or carditis, alone or in combination.
 - Subcutaneous nodules and erythema marginatum (skin rash) may be involved.

- These symptoms usually improved but can cause persistent damage to one or more the heart valves {Rheumatic Heart Disease (RHD)}, especially with repeated infection.
- Chronic rheumatic fever, generally defined as disease persisting for longer than 6 months, occurs in less than 5% of cases.

Epidemiology

- The incidence of ARF is 19 cases/100 000 school- aged children worldwide, but 2 to 14 cases/100 000 in the United States.
- About 35 million people worldwide have rheumatic heart disease. Most cases occur in low-and middle-income countries.
- Decrease incidence of rheumatic fever in the developed countries due to improved hygiene and use of antibiotics.
- ARF can occur at any age, mostly affect children and adolescents 5 to 15 year of age, rare before age 4 and uncommon after 18.

Causes and pathogenesis

- The cause and pathogenesis of ARF is **still not fully understood**
- It may be develops due to:
 - An abnormal autoimmune reaction triggered by molecular mimicry between the cell-wall M proteins of infecting *Streptococcus pyogenes* and certain host tissues as cardiac myosin.

Risk factors

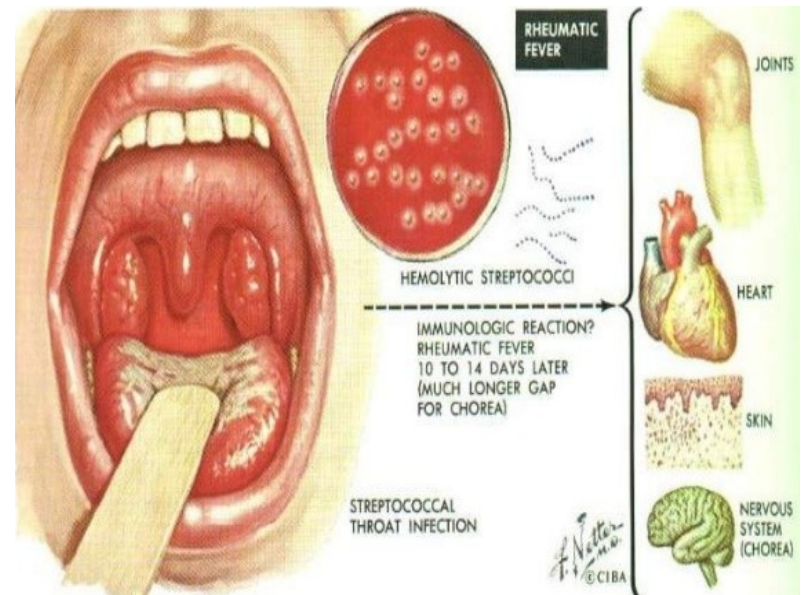


- Aged between 5-15 years are most at risk of getting rheumatic fever, especially those with one or more of the following risk factors:
- Familial: family history of rheumatic fever, rheumatic heart disease or both
- Malnutrition.
- Overcrowding.
- Low socioeconomic status predispose infections and subsequent rheumatic attack



Clinical Pictures

- **Acute rheumatic fever:**
 - The clinical features starts 2–3 (5) weeks after streptococcal infection.
 - Fever, malaise, tiredness, arthralgia (joint pain), sore throat, abdominal pain.



- **Clinical manifestations (major):**

- **Arthritis:**

- It is a major manifestation in 92% of adults.
 - Most common and **earliest manifestation.**
 - **Character:**
 - **Polyarthritis** (more than 4 joint).
 - **Affect large joints** as knees, ankles, elbows, and wrists **causing** limitation of movements.
 - **Fleeting, migratory, asymmetrical.**
 - **Monoarthritis** may occur.
 - It subsides within 4-6 weeks without any permanent damage.
 - Treatment: **Aspirin (NSAIDs).**

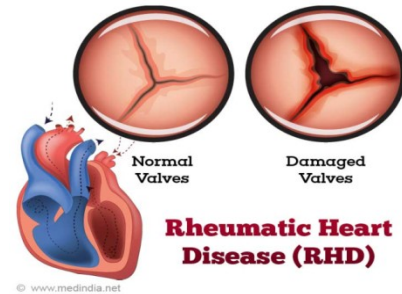


- **Pancarditis (30-60%):**

- It is more common in young children but can occur in adults.

- **Clinical pictures:**

- **Chest pain.**
 - **Dyspnea:** difficulty of breathing or breathlessness
 - Lower limb edema.
 - Palpitation (Uncomfortable awareness of heart beats). (arrhythmia).
 - **Other features:** tachycardia and new or changed murmurs (mitral, aortic area).



- **Diagnosis:** clinical examination, echo.

- **Treatment:**

- » **Corticosteroids:** Prednisolone. After 2-3 weeks, the dosage may be tapered, reduced by 25% each week.
 - » **Heart failure.**

- **Sydenham's chorea (10-30%):**

- It occurs **later (6 months or more)** after initial infection, so ESR, CRP and ASO titers were normal.

- It **affects females**, particularly in children and adolescence, **very rare in adults**.

- **Clinical pictures:**

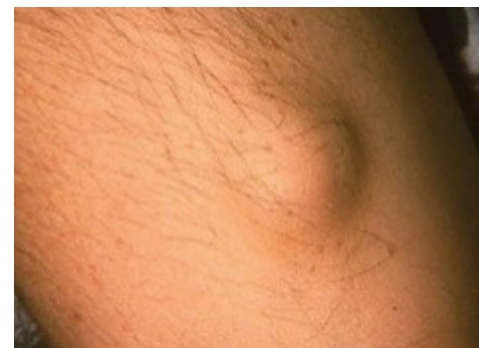


- **Purposeless movements affect arms, face (tongue) and legs (unilateral or bilateral). It disappears during sleep.**

- It usually **resolves without permanent damage** but occasionally **lasts 2-3 years**.

- **It is a major problem** for the patient and her family.

- **Subcutaneous nodules (0-8%):**
 - Uncommon, painless, firm and small (0.5-2 cm).
 - It associated with severe carditis.
 - Its present **extensor surfaces** of large joints, over bony prominence.
 - It **appear 1–2 weeks after other symptoms**, last 1-2 weeks (rarely more than 1 month). It may **persist intermittently for weeks to months**.

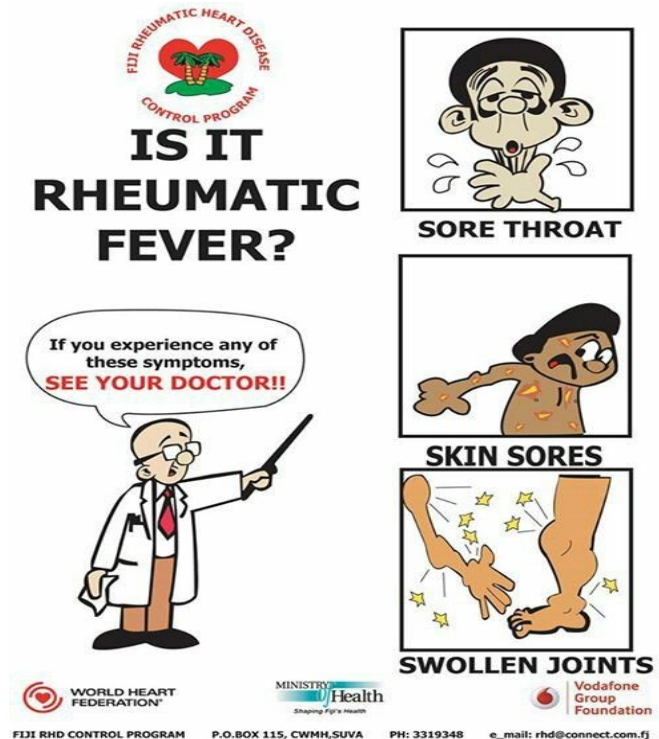


- **Erythema marginatum (less 2%):**
 - Highly specific.
 - Transient pink rash with slightly raised edges and occur on the **trunk and proximal extremities** and almost never on the face.
 - Non-scarring, not **painful or itchy rash**.



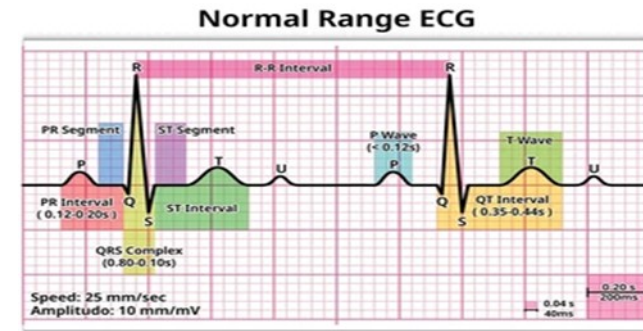
Investigations

- No single laboratory test can confirm diagnosis of ARF.
- Complete cell count (CBC): high WBC (12,000-20,000/ μ L)
- Elevated ESR and CRP
- Positive throat swab culture for group A β -hemolytic streptococci.
- Blood cultures are obtained to rule out infective endocarditis.



- **Antistreptolysin O titer (ASOT):**
 - High titer used to diagnose **streptococcal infection**.
 - A positive test usually is > 200 units/mL, but normal ranges vary from laboratory to laboratory and by age.

- **Electrocardiogram (ECG):**
abnormalities as prolonged PR interval.
- **Plain X ray Chest:** cardiomegaly and pulmonary congestion.
- **Echocardiography:** cardiac dilatation and valve abnormalities.

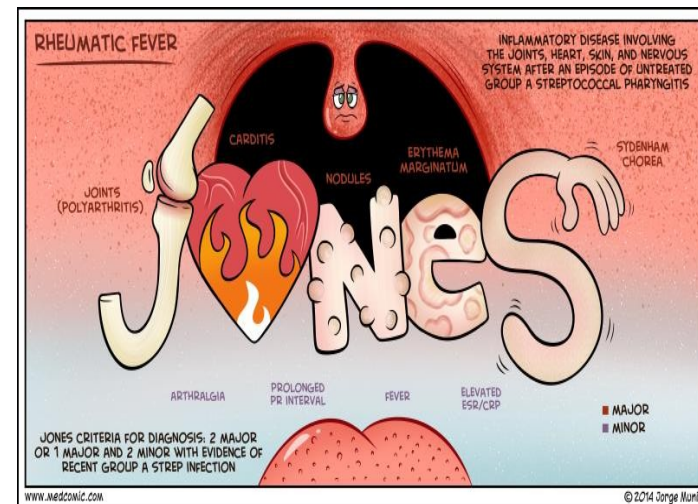


• Diagnosis of ARF:

- Presence of two or more major criteria, or
- One major plus two or more minor criteria with
- Evidence of streptococcal infection:
 - Positive throat culture for group A streptococcus.
 - History (e.g. of scarlet fever).
 - Elevated antistreptolysin O titre (or other serological assay for streptococci).

Rheumatic Fever: Criteria		knowmedge
Mnemonic: "JONES CAFE PAL"		
Major Criteria		Minor Criteria
J	Joint Involvement	C CRP Increased
O	O looks like a heart = myocarditis	A Arthralgia
N	Nodules, subcutaneous	F Fever
E	Erythema marginatum	E Elevated ESR
S	Sydenham chorea	P Prolonged PR Interval
		A Anamnesis of Rheumatism
		L Leukocytosis

Diagnosis	
Throat cultures growing GABHS OR Elevated anti-streptolysin O titers	2 Major criteria OR 1 Major criterion and 2 Minor criteria



Treatment



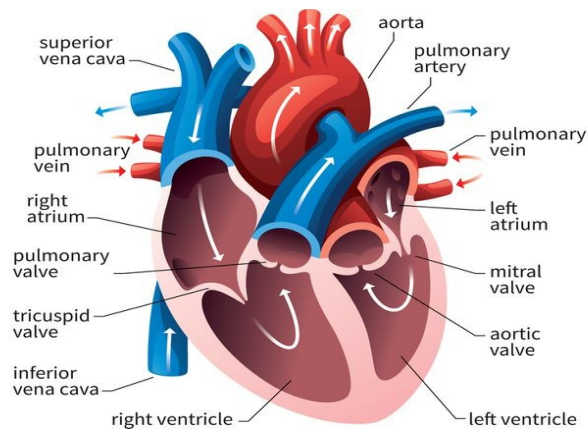
- **Treatment strategies for ARF include the following:**
 - **Management of the acute attack**
 - **Management of the current infection**
 - **Prevention of further infection and attacks**

Management of the acute attack and current infection

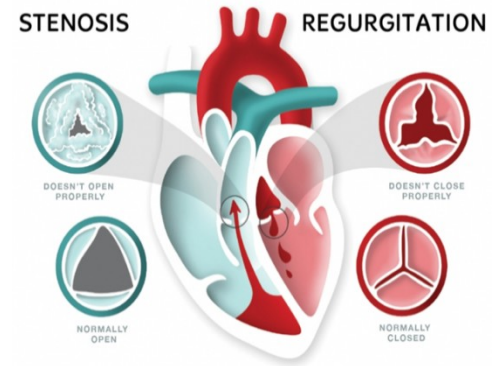
- Bed rest (carditis) and supportive therapy.
- **Antibiotics** as Procaine penicillin.
 - Erythromycin or clarithromycin is used if the patient is allergic to penicillin
- **Anti-inflammatory** agents as Aspirin (arthritis, fever, and other acute symptoms).
- Prednisone in moderate to severe carditis.
- **Surgical** to treat long-term valvular damage from RHD.
 - Valve repair or replacement.

- **Treat complication as heart failure, arrhythmia.**
- **Symptomatic treatment of Sydenham chorea, including:**
 - **Anticonvulsants (e.g., valproate, carbamazepine)**
 - **Neuroleptics (e.g., haloperidol).**
 - **Gradual withdrawal of medication is suggested once patients have been symptom free for at least 1 month.**

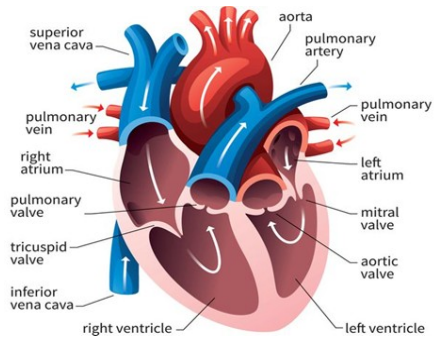
- **Prophylactic:**
 - **Primary:** single intramuscular (IM) injection of benzathine benzylpenicillin.
 - **Secondary prevention:**
 - The American Heart Association (AHA) Committee on Acute Rheumatic Fever recommends a regimen consisting of **benzathine benzylpenicillin at 1.2 million units IM/ 3-4 weeks.**
 - Benzathine penicillin G monthly IM injection until the age of 20 years or at least 10 years after the latest attack.
 - Oral cephalosporins, Erythromycin or clarithromycin is used if the patient is allergic to penicillin



Prognosis



- More than 50% of those who suffer acute rheumatic fever with carditis will later (after 10–20 years) **develop chronic rheumatic valvular disease**, predominantly **affecting mitral and aortic valves**; tricuspid valve is rarely affected, and the pulmonary valve is hardly ever involved.
 - It is chronic and progressive.
 - **Mitral valve stenosis (MS)** commonly in adults, most common complication is atrial fibrillation.
 - **Mitral regurgitation (MR)** most commonly in children and adolescent.



Valvular Heart Disease
MITRAL STENOSIS (MS)

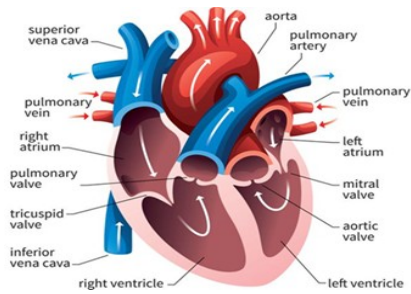
- It is a common valvular lesion in RHD
- When the normal valve orifice area is reduced to 1 cm², severe mitral stenosis is present.
- To maintain cardiac output, the **left atrial pressure increases** and left **atrial hypertrophy and dilatation** occur.
- Consequently, **pulmonary venous**, pulmonary arterial and right heart pressures also increase.
- **Pulmonary hypertension** leads to right ventricular hypertrophy, dilatation and failure.

- **Symptoms:**

- Usually, **asymptomatic**.
- Progressively severe dyspnea develop due to **pulmonary venous** hypertension and recurrent bronchitis.
- Productive **cough with blood-tinged**, frothy sputum and occasionally frank hemoptysis may occur.
- **Right heart failure** (weakness, fatigue and abdominal or lower limb edema).
- Palpitations (Atrial fibrillation).
- Systemic emboli (Atrial fibrillation).

COMPLICATIONS OF MITRAL STENOSIS

- **Atrial fibrillation**
- **Systemic embolization**
- **Pulmonary hypertension**
- **Pulmonary infarction**
- **Chest infections**
- **Infective endocarditis (rare)**
- **Tricuspid regurgitation**
- **Right ventricular failure**

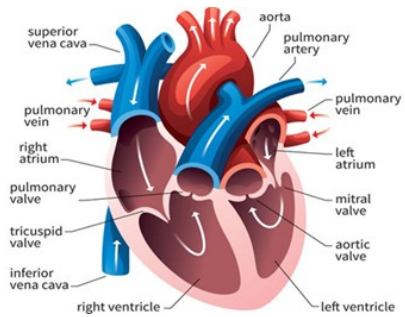


Mitral Regurgitation

- **Pathophysiology:**

- **Regurgitation into the left atrium** produces left atrial dilatation but little increase in left atrial pressure.
- Since a proportion of the stroke volume is regurgitated, the stroke volume increases to maintain the forward cardiac output and the **left ventricle therefore enlarges.**

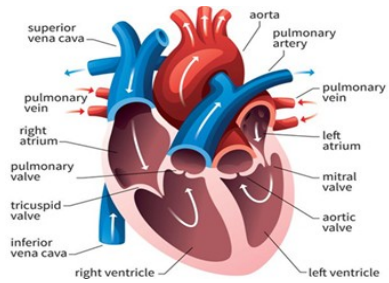




Aortic stenosis

- **Pathophysiology:**

- Obstructed left ventricular emptying leads to increased **left ventricular pressure** and compensatory **left ventricular hypertrophy**.
- In turn, this results in relative **ischemia of the left ventricular myocardium**, and consequent **angina**, **arrhythmias** and **left ventricular failure**.



Aortic Regurgitation

- **Symptoms:**

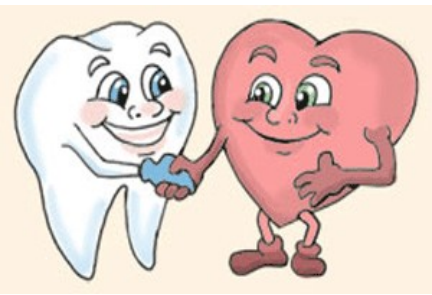
- In aortic regurgitation, significant **symptoms occur late** and do not develop until **left ventricular failure** occurs.
- As with mitral regurgitation, a common symptom is 'pounding of the heart' because of the increased left ventricular size and its vigorous pulsation.
- **Angina pectoris** is a frequent complaint.
- Varying grades **of dyspnea** occur depending on the extent of left ventricular dilatation and dysfunction.
- Arrhythmias are relatively uncommon.



Dental care



- **Routine and regular dental care is important for ARF/ RHD to avoid risk of infective endocarditis.**
 - Some dental procedures do have an increased risk of bacteremia.
- **All patients should receive education about oral hygiene, and should be referred for dental assessment and treatment when required.**



- **Antibiotic prophylaxis** is recommended for at risk patients during dental procedures.
- Most patients with RHD are anticoagulated; dental treatment can be done **under LA** in consultation with the physician.
- **Conscious sedation with nitrous oxide** may be given if cardiac function is good and with the approval of the physician.



Thank you!!!

